

Regression Analysis Step-by-Step

MDragon Data Tools - Comprehensive Guide

The Complete Process

Step 1: Collect and Examine Data

- Plot scatter diagram
- Check for linear relationship
- Identify outliers

Step 2: Calculate Regression Equation

1. Find means: $\bar{x}$, $\bar{y}$
2. Calculate b_1 (slope)
3. Calculate b_0 (intercept)
4. Form equation: $\hat{y} = b_0 + b_1x$

Step 3: Assess Model Fit

- Calculate r (correlation)
- Calculate R^2 (coefficient of determination)
- Examine residual plots

Step 4: Test Significance

- Test $H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$
- Use t-test or ANOVA F-test

Step 5: Make Predictions

- Interpolation (within data range): reliable
- Extrapolation (outside data range): risky

Detailed Example

Data: Advertising spend (X, \$1000s) vs Sales (Y, \$1000s)

X: 1, 2, 3, 4, 5

Y: 10, 15, 18, 22, 26

Calculations

1. Means:

$$\bar{x} = 3, \bar{y} = 18.2$$

2. Slope:

$$b_1 = 72/10 = 3.6$$

3. Intercept:

$$b_0 = 18.2 - 3.6(3) = 7.4$$

4. Equation:

$$\hat{y} = 7.4 + 3.6x$$

5. Correlation:

$r = 0.985$ (strong positive)
 $R^2 = 0.97$ (97% of variance explained)

Interpretation:

- Each \$1,000 increase in advertising → \$3,600 increase in sales
- Model explains 97% of sales variation
- Very strong linear relationship

Master regression with our interactive tools!

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mondragon-developer.github.io/statools